

Khám phá Dữ liệu

Analysis: Data Exploration

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 07

Góc nhìn Chuyên gia (Professional Insight)

- **Tầm quan trọng:** Khám phá dữ liệu là bước đầu tiên để "làm quen" với dữ liệu. Kế toán viên cần hiểu rõ dữ liệu của mình trước khi đưa ra bất kỳ kết luận hay báo cáo nào.
- **Trực giác Kế toán:** Kết hợp kinh nghiệm thực tiễn và kỹ năng phân tích để phát hiện ra các điểm bất thường, xu hướng hoặc insight tiềm ẩn.
- **Kể chuyện bằng dữ liệu (Data Storytelling):** Một biểu đồ trực quan tốt có giá trị hơn hàng ngàn con số trong bảng tính phức tạp.

Lộ trình Chương (Chapter Roadmap)

- **LO 7.1:** Quy trình Khám phá Dữ liệu.
- **LO 7.2:** Khám phá Mọi quan hệ Nền tảng qua 8 Mẫu Trực quan hóa.
- **LO 7.3:** Khám phá Dữ liệu bằng cách Tích hợp Mọi quan hệ (Dashboards).

7.1 Khám phá Dữ liệu (Data Exploration) là gì?

- **Định nghĩa:** Là quá trình phân tích dữ liệu ban đầu bằng các công cụ trực quan và thống kê để hiểu rõ đặc điểm, phát hiện xu hướng và kiểm tra các giả định.
- **Sự khác biệt:**
 - **Khám phá (Exploration):** Tự do tìm kiếm insight (cho chính bạn).
 - **Diễn giải (Interpretation):** Hiểu ý nghĩa của insight.
 - **Báo cáo (Reporting):** Trình bày insight cho người khác (Management/Stakeholders).

ILLUSTRATION 7.1

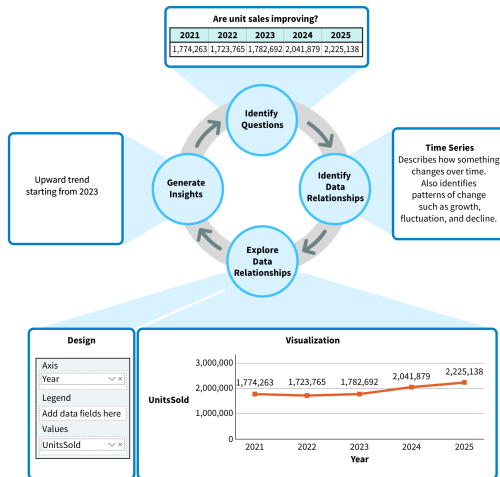


ILLUSTRATION 7.2

Country	Model	Type	2021	2022	2023	2024	2025
USA	Civic	Sedan	320981	301882	292331	291002	255423
USA	Accord	Sedan	243192	245998	231441	309885	344771
USA	CR-V	SUV	175883	160886	190001	220877	252019
USA	Pilot	SUV	140444	142980	139441	142917	125090
USA	Odyssey	Minivan	188664	167123	150872	150881	139009
USA	Ridgeline	Truck	58322	55897	56889	55899	54891
Canada	Civic	Sedan	230887	242998	275667	381998	480871
Canada	Accord	Sedan	195232	200872	210665	253988	319755
Canada	CR-V	SUV	135423	129809	126592	114119	98077
Canada	Pilot	SUV	40998	35672	39811	43125	47329
Canada	Odyssey	Minivan	25229	7761	26981	22099	19822
Canada	Ridgeline	Truck	19008	31887	42001	55089	88081

(A) Cross Tabulation Structure of HNA Data Set

Country	Model	Type	Year	UnitsSold
Canada	Accord	Sedan	2021	195232
Canada	Accord	Sedan	2022	200872
Canada	Accord	Sedan	2023	210665
Canada	Accord	Sedan	2024	253988
Canada	Accord	Sedan	2025	319755
Canada	Civic	Sedan	2021	230887
Canada	Civic	Sedan	2022	242998
Canada	Civic	Sedan	2023	275667
Canada	Civic	Sedan	2024	381998
Canada	Civic	Sedan	2025	480871

(B) Flat Structure of HNA Data Set

ILLUSTRATION 7.3

Models	TotalUnitsSold
Civic	3,074,040
Accord	2,555,799
CR-V	1,603,686
Odyssey	898,441
Pilot	897,807
Ridgeline	517,964
TotalUnitsSold	9,547,737

(A) PivotTable: Total Units Sold Per Model⁴

PivotTable Fields

Choose fields to add to report:

Search

☐ Country
☒ **Model**
☐ Type
☐ Year
☒ **UnitsSold**
More Tables..

Drag fields between areas below:

Filters Columns

Rows Values

Model TotalUnitsSold

☐ Defer Layout Update Update

(B) PivotTable Creation

ILLUSTRATION 7.4

Total UnitsSold Models ▼	Countries ⌵ Canada	USA	TotalUnitsSold
Accord	1,180,512	1,375,287	2,555,799
Civic	1,612,421	1,461,619	3,074,040
CR-V	604,020	999,666	1,603,686
Odyssey	101,892	796,549	898,441
Pilot	206,935	690,872	897,807
Ridgeline	236,066	281,898	517,964
TotalUnitsSold	3,941,846	5,605,891	9,547,737

ILLUSTRATION 7.5

Filters	Columns
Country ▼	
Rows	Σ Values
Model ▼	TotalUnitsSold ▼

(A) Creating a PivotTable Filter

Country	USA
Models ▼	TotalUnitsSold
Accord	1,375,287
Civic	1,461,619
CR-V	999,666
Odyssey	796,549
Pilot	690,872
Ridgeline	281,898
TotalUnitsSold	5,605,891

(B) PivotTable with Filter Applied

ILLUSTRATION 7.6

Model	TotalUnitsSold
Accord	2,555,799
Civic	3,074,040
CR-V	1,603,686
Odyssey	898,441
Pilot	897,807
Ridgeline	517,964
Total	9,547,737

(A) Power BI Matrix

The screenshot illustrates the Power BI interface for creating a matrix. The interface is divided into three main sections: Filters (left), Visualizations (center), and Fields (right). The Visualizations pane shows the Matrix icon selected, with 'Model' in the Rows field and 'TotalUnitsSold' in the Values field. The Fields pane shows the 'UNITSALES' table expanded with 'Model' and 'UnitsSold' selected. The Filters pane shows 'Model' and 'TotalUnitsSold' as filters on the visual. The interface is annotated with red boxes and numbers 1 through 6.

- 1: Matrix visualization icon in the Visualizations pane.
- 2: Fields pane.
- 3: Rows field containing 'Model'.
- 4: Columns field containing 'Add data fields here'.
- 5: Values field containing 'TotalUnitsSold'.
- 6: Filters pane.

(B) Power BI Matrix Creation

Quy trình Khám phá Dữ liệu

- **Bước 1: Xác định Mục tiêu (Identify Objectives):** Bạn đang tìm kiếm điều gì?
- **Bước 2: Lựa chọn Biến (Select Variables):** Dữ liệu nào cần thiết?
- **Bước 3: Khám phá Trực quan (Visual Exploration):** Sử dụng biểu đồ.
- **Bước 4: Đánh giá & Tinh chỉnh (Evaluate & Refine):** Biểu đồ có ý nghĩa không? Có cần thay đổi không?

ILLUSTRATION 7.7

**Breakdown of Total Units Sold by
Model: 2021-2025**

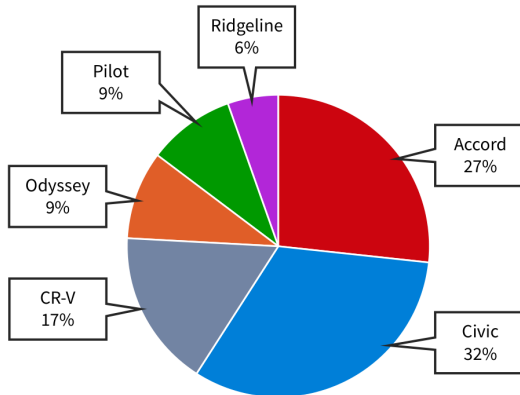


ILLUSTRATION 7.8

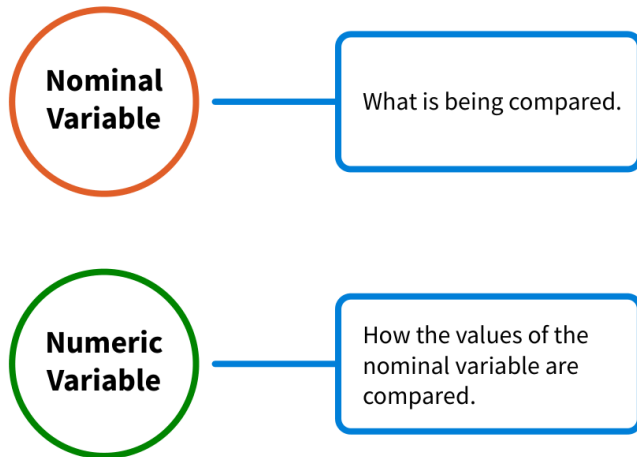


ILLUSTRATION 7.9

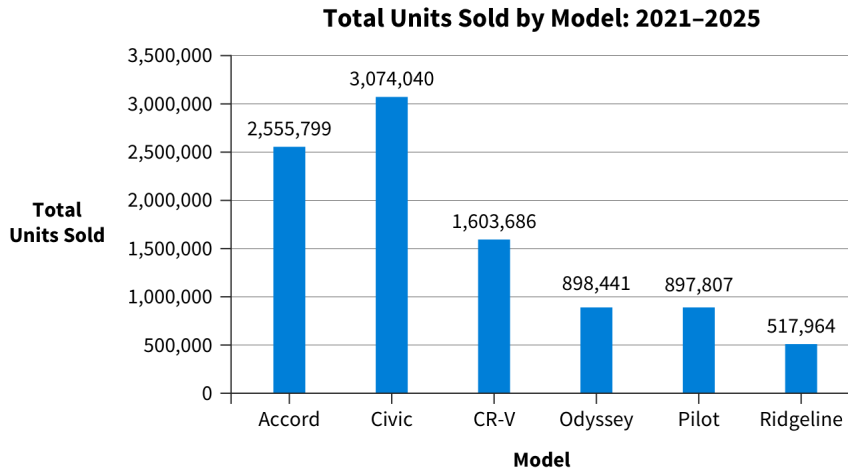


ILLUSTRATION 7.10

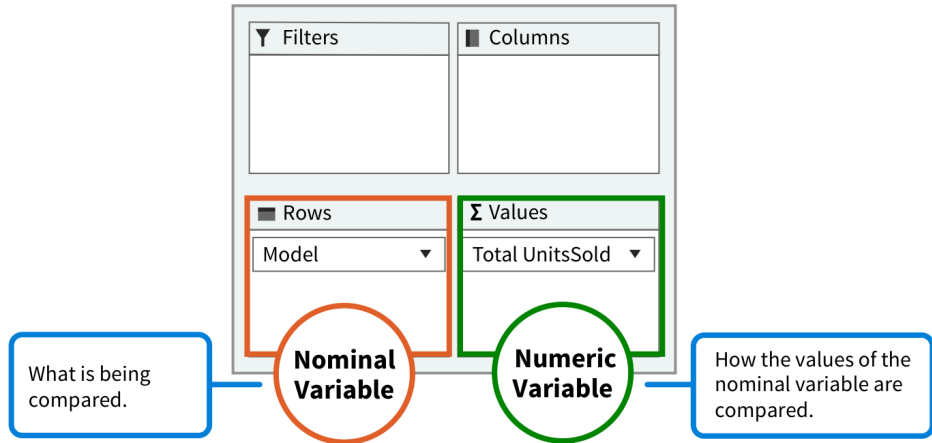


ILLUSTRATION 7.11

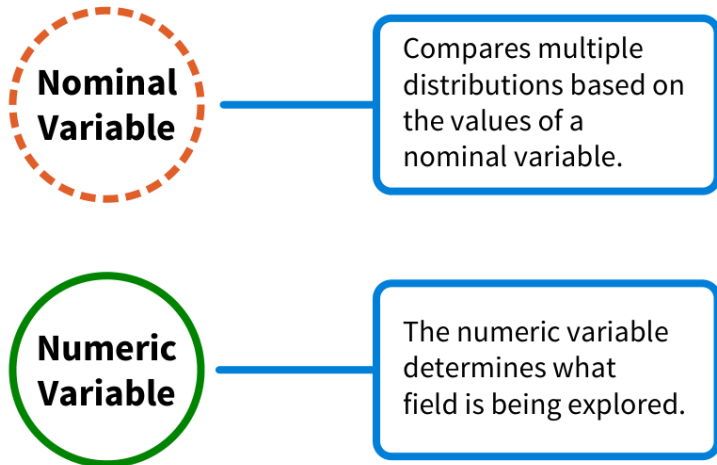


ILLUSTRATION 7.12

Field Name	Description
ID	The internal ID that identifies the car.
Model	The car's model.
Profit	The profit made when selling the car is selling price minus dealer invoice. This is also known as front-end gross profit.

- **PivotTables trong Excel:** Công cụ cơ bản, nhanh chóng và phổ biến nhất để nhóm, tóm tắt và lọc dữ liệu.
- **Đa nền tảng (Across Tools):** Các công cụ BI hiện đại (Power BI, Tableau) cung cấp khả năng trực quan hóa mạnh mẽ hơn, hỗ trợ tương tác và xử lý dữ liệu lớn (Big Data).
- **Mục tiêu cốt lõi:** Tìm ra các *Insights* ẩn giấu đằng sau những con số tài chính khô khan.

ILLUSTRATION 7.13

ID	Model	Profit
1	Civic	\$ 2,125
2	CR-V	\$ 2,798
3	Odyssey	\$ 2,144
4	Accord	\$ 2,749
5	Accord	\$ 1,810
6	Accord	\$ 1,844
7	Odyssey	\$ 4,446
8	Civic	\$ 1,912
9	Pilot	\$ 3,107
10	Civic	\$ 2,101

ILLUSTRATION 7.14

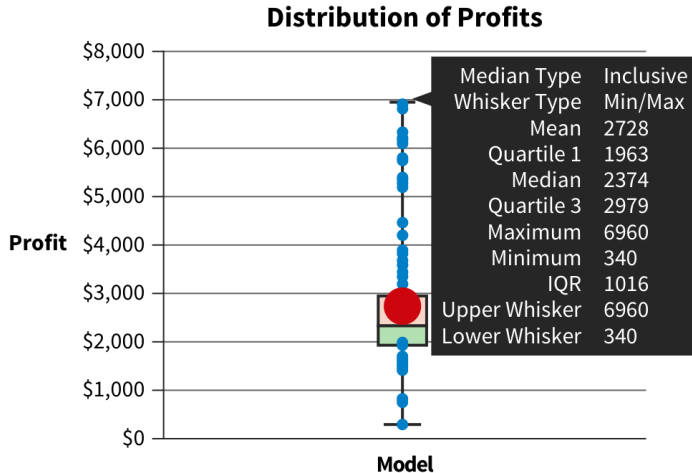


ILLUSTRATION 7.15

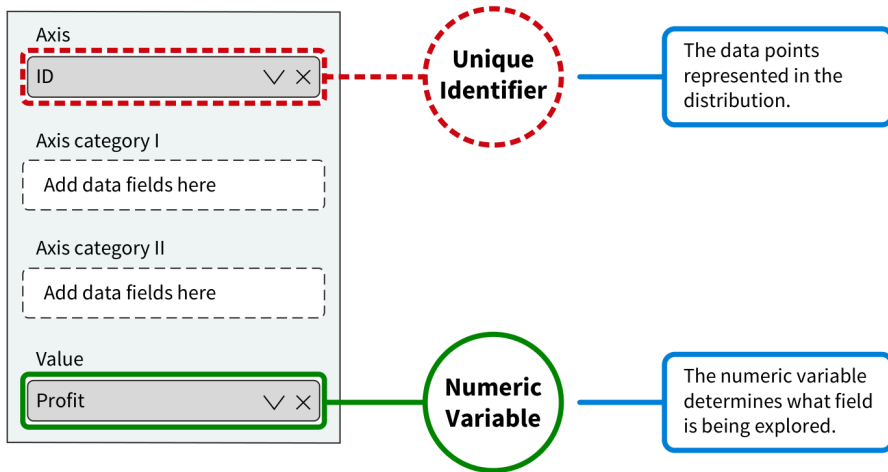


ILLUSTRATION 7.16

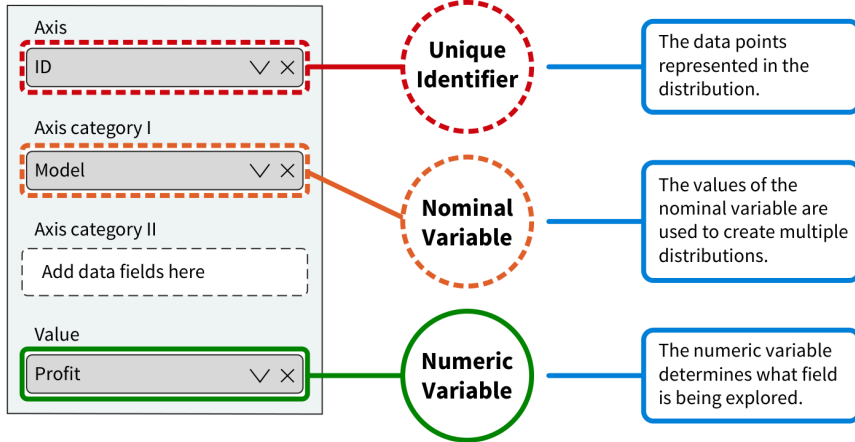
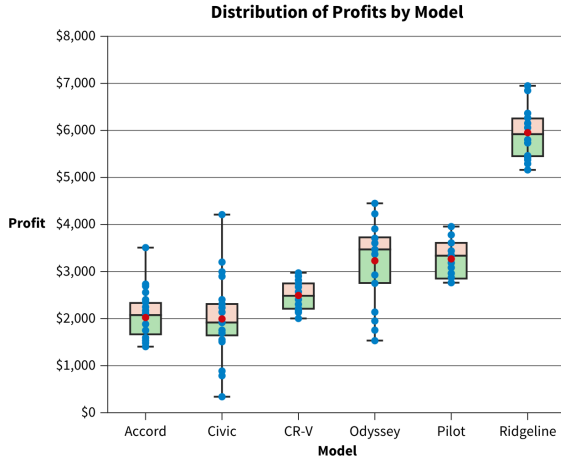


ILLUSTRATION 7.17



7.2 Mỗi quan hệ Nền tảng qua Trực quan hóa

- **Mục đích:** Giúp não bộ con người nhanh chóng nhận diện quy luật (Pattern Recognition).
- **8 Mẫu Mỗi quan hệ Nền tảng (Eight Patterns):**
 - 1 Phần-Toàn thể (Part-To-Whole)
 - 2 So sánh Cường độ (Magnitude)
 - 3 Chuỗi Thời gian (Time Series)
 - 4 Phân phối (Distribution)
 - 5 ... và 4 mẫu nâng cao khác.

ILLUSTRATION 7.18

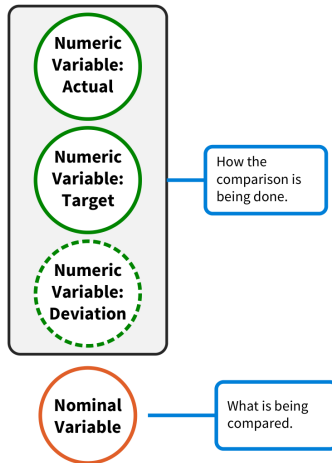


ILLUSTRATION 7.19

Country	Model	Type	ActualUnits	BudgetedUnits
US	Civic	Sedan	255423	300000
US	Accord	Sedan	344771	350000
US	CR-V	SUV	252019	25000
USA	Pilot	SUV	125090	160000
USA	Odyssey	Minivan	139009	145000
USA	Ridgeline	Truck	54891	55000
Canada	Civic	Sedan	480871	425000
Canada	Accord	Sedan	319755	275000
Canada	CR-V	SUV	98077	110000
Canada	Pilot	SUV	47329	45000
Canada	Odyssey	Minivan	19822	30000
Canada	Ridgeline	Truck	88081	65000

ILLUSTRATION 7.20

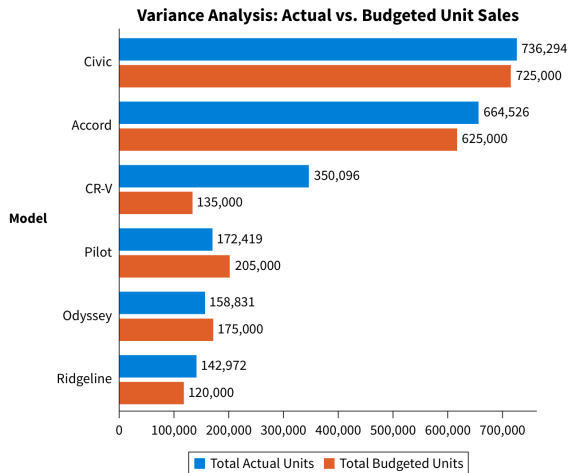


ILLUSTRATION 7.21

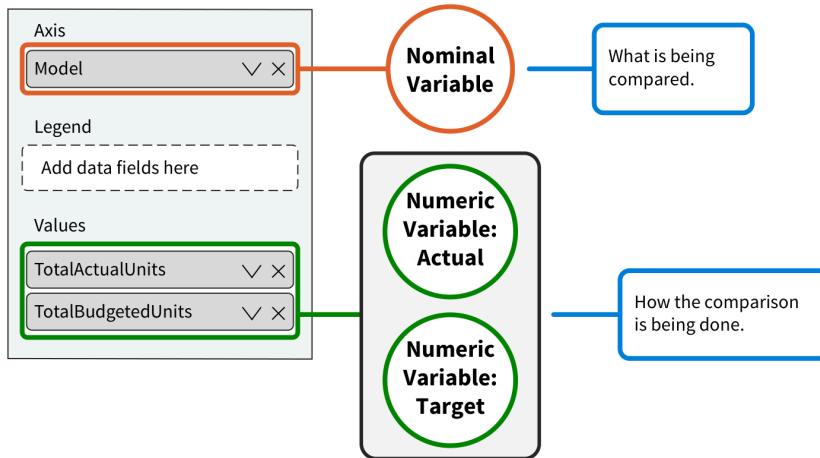
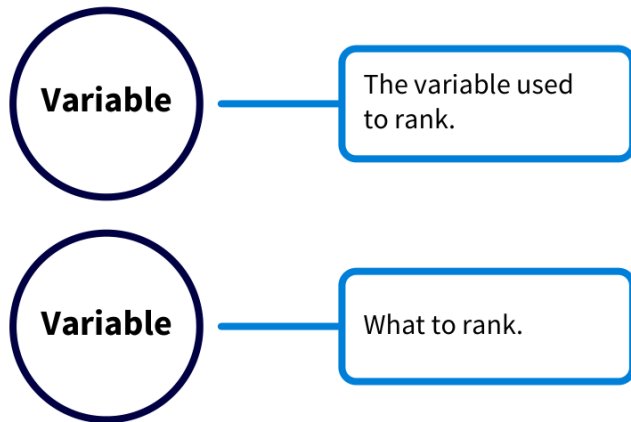


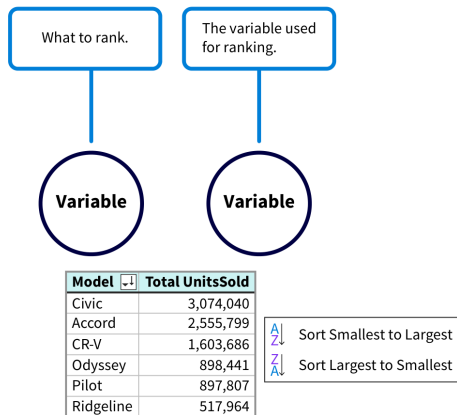
ILLUSTRATION 7.22



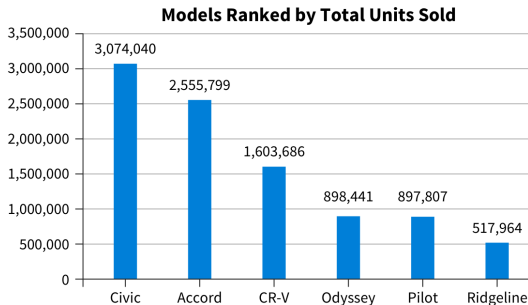
Phân tích Các Mẫu Cơ bản

- **Phần-Toàn thể (Part-To-Whole):** Biểu đồ tròn (Pie), Biểu đồ vành khăn (Donut), Treemap. Cho thấy tỷ trọng đóng góp (Ví dụ: Tỷ trọng doanh thu theo từng sản phẩm).
- **Cường độ (Magnitude):** Biểu đồ cột (Bar/Column). Dùng để so sánh trực diện kích thước, quy mô giữa các hạng mục.
- **Chuỗi Thời gian (Time Series):** Biểu đồ đường (Line). Thể hiện xu hướng biến động qua thời gian (Tháng, Quý, Năm).

ILLUSTRATION 7.23



(A) Creating a Ranking Relationship for a PivotTable



(B) Integrating a Ranking Relationship with a Column Chart

ILLUSTRATION 7.24

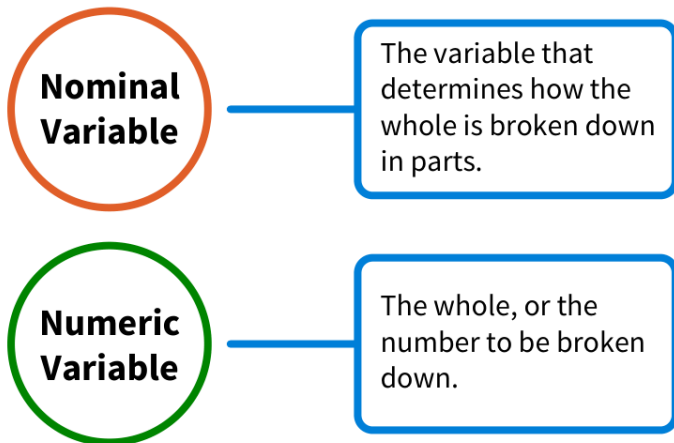


ILLUSTRATION 7.25

Breakdown of Units Sold by Model: 2021–2025

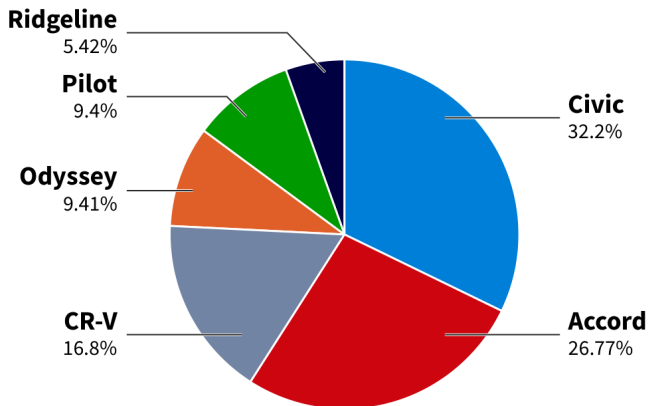
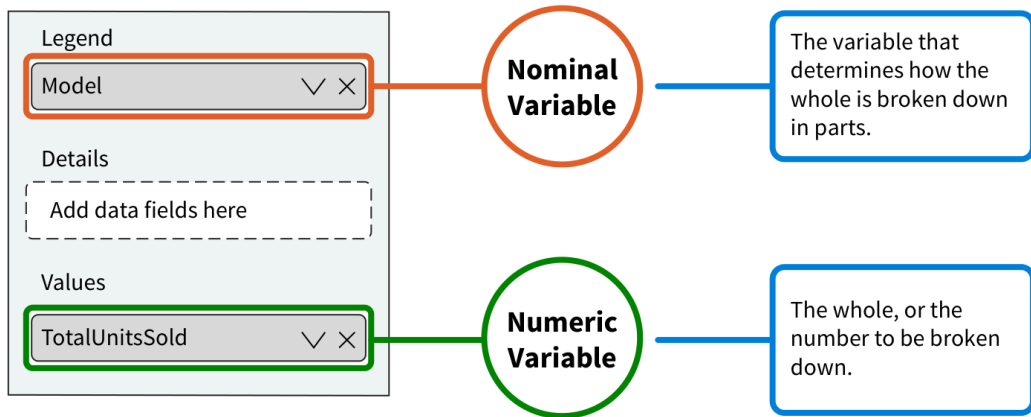


ILLUSTRATION 7.26



Phân tích Các Mẫu Nâng cao

- **Phân phối (Distribution):** Biểu đồ Histogram, Box-plot. Đánh giá sự phân tán của dữ liệu và phát hiện ngoại lai (Outliers).
- **Tương quan (Correlation):** Biểu đồ phân tán (Scatter plot). Kiểm tra mối liên hệ giữa hai biến (Ví dụ: Chi phí quảng cáo và Doanh thu).
- **Không gian/Địa lý (Spatial):** Bản đồ (Map). Hiển thị dữ liệu theo vị trí địa lý (Ví dụ: Doanh số theo Tỉnh/Thành phố).

ILLUSTRATION 7.27

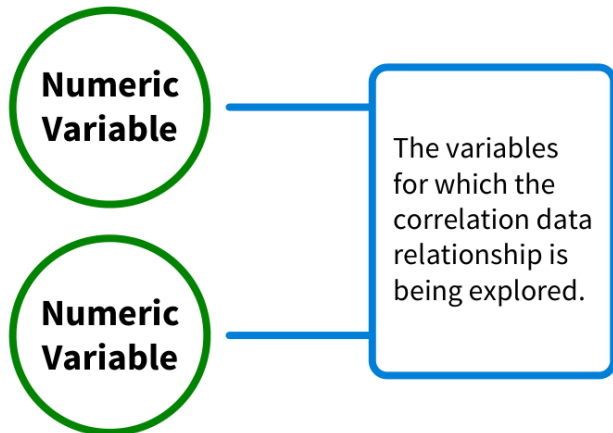


ILLUSTRATION 7.28

Property	Size	Time
1	4500	9
2	3000	6
3	50000	50
4	25000	45
5	4000	8
6	18000	18
7	21000	17
8	100000	120
9	22000	55
10	23000	25

ILLUSTRATION 7.29

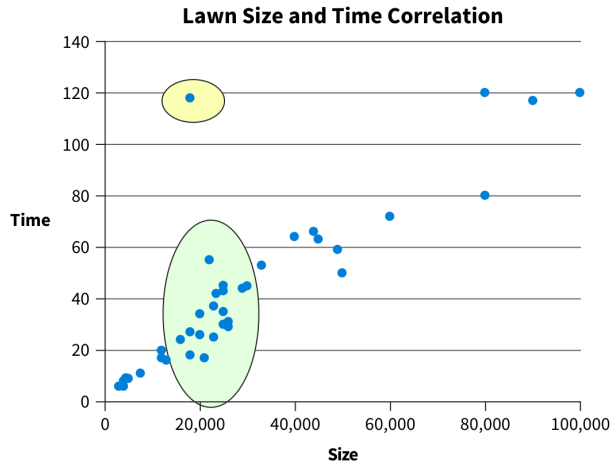
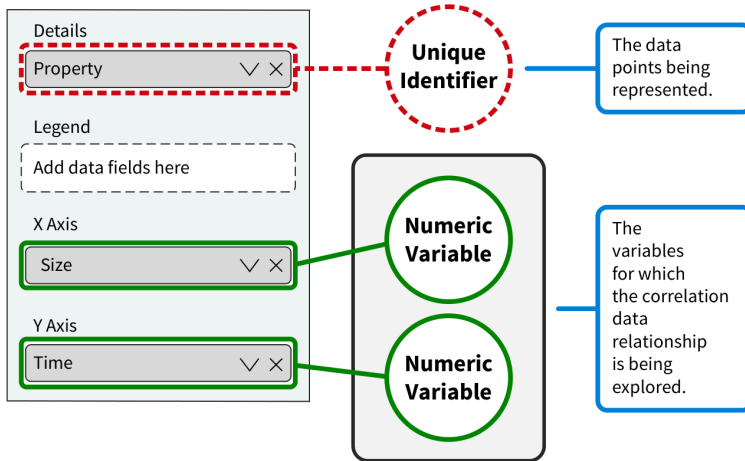


ILLUSTRATION 7.30



Nguyên tắc Thiết kế Biểu đồ

- **Giữ sự đơn giản (Keep it Simple):** Tránh lạm dụng 3D, bóng đổ hoặc màu sắc lòe loẹt gây xao nhãng (Chartjunk).
- **Rõ ràng (Clarity):** Trục tọa độ, nhãn dữ liệu (Data labels), và tiêu đề phải dễ đọc và mang tính mô tả.
- **Sử dụng Màu sắc có chủ ý:** Dùng màu đỏ cho sự giảm sút/rủ ro, màu xanh cho sự tăng trưởng.

ILLUSTRATION 7.31

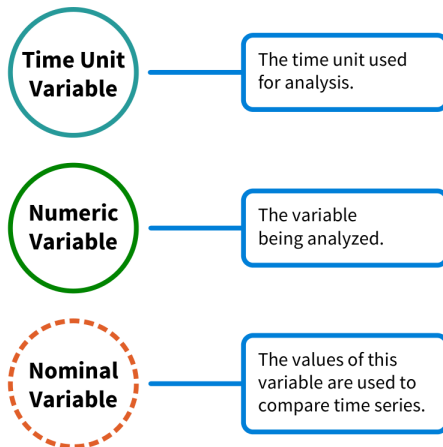


ILLUSTRATION 7.32

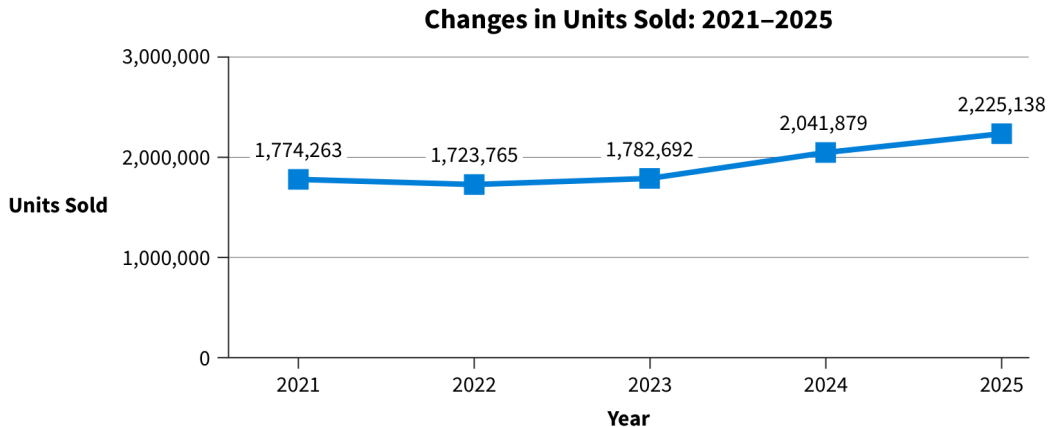


ILLUSTRATION 7.33

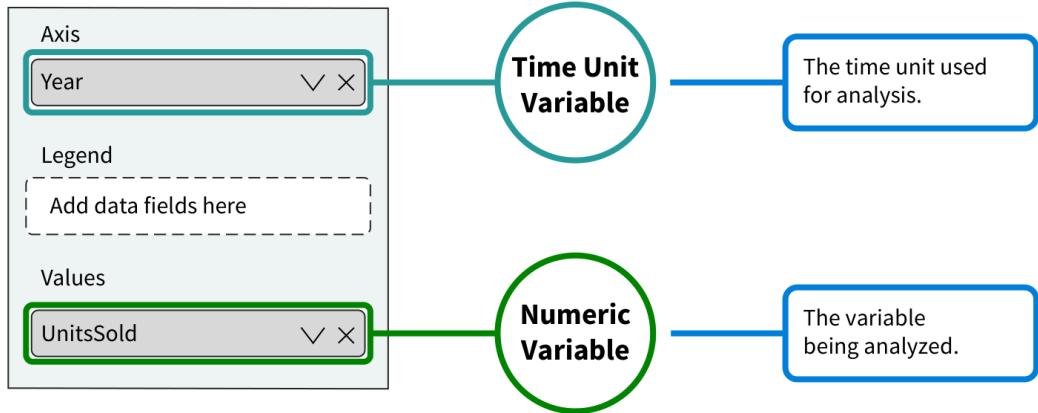
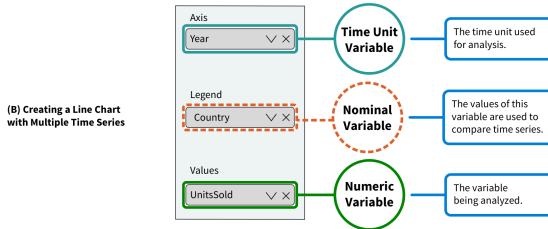
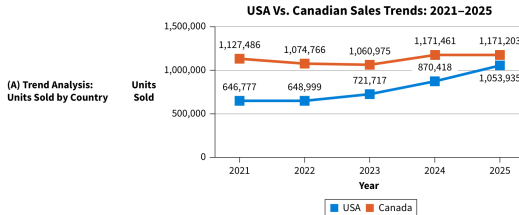


ILLUSTRATION 7.34



7.3 Khám phá Dữ liệu Đa chiều (Tích hợp)

- **Tích hợp Mỗi quan hệ:** Sử dụng kết hợp nhiều loại biểu đồ để có cái nhìn toàn diện hơn (Holistic view).
- **Hai mẫu trong một (Single Visualization):**
 - Biểu đồ Đa trục (Dual-axis): So sánh hai chỉ số có thang đo khác nhau trên cùng một biểu đồ (ví dụ: Doanh thu - Cột, Tỷ suất LN - Đường).
 - Biểu đồ Bong bóng (Bubble chart): Thể hiện 3 biến dữ liệu (Trục X, Trục Y, Độ lớn bong bóng).

ILLUSTRATION 7.35

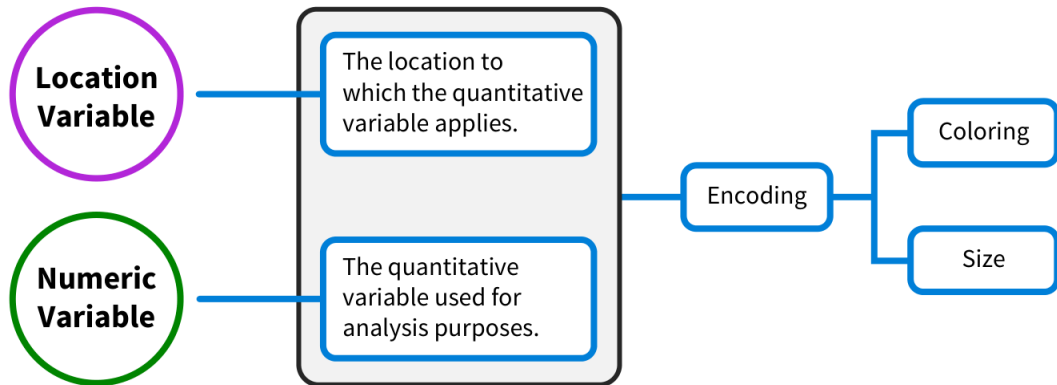


ILLUSTRATION 7.36

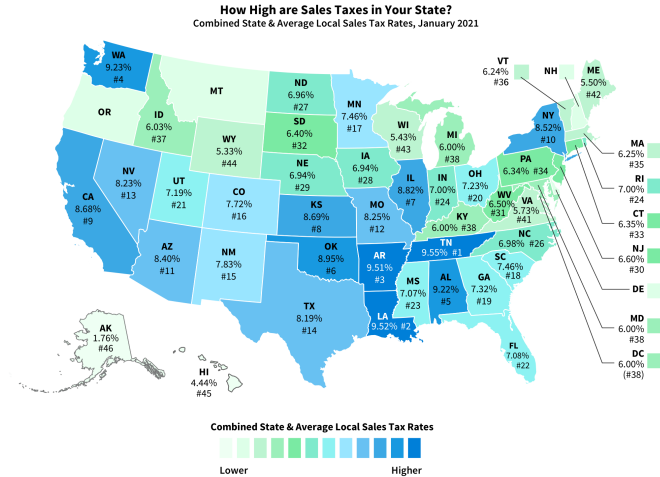


ILLUSTRATION 7.37

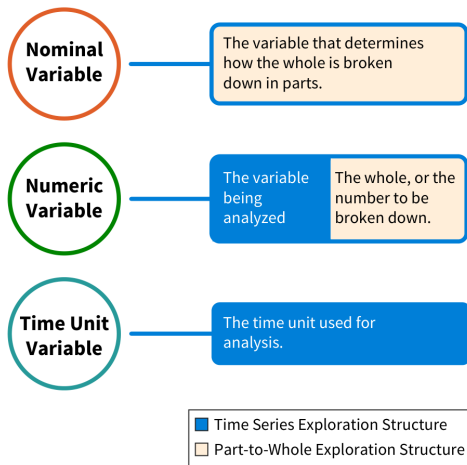


ILLUSTRATION 7.38

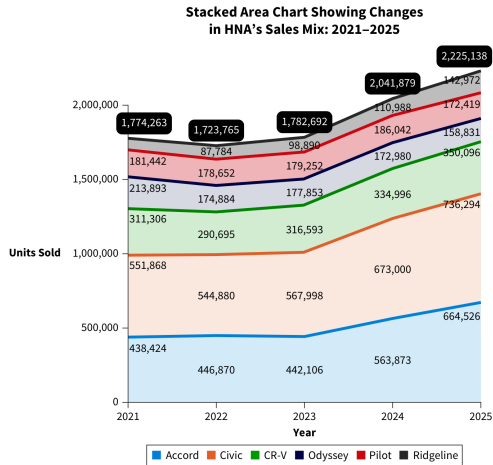


ILLUSTRATION 7.39

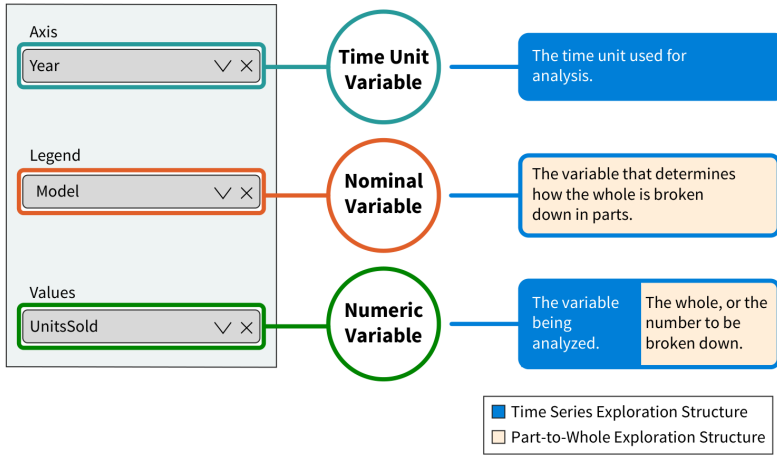


ILLUSTRATION 7.40

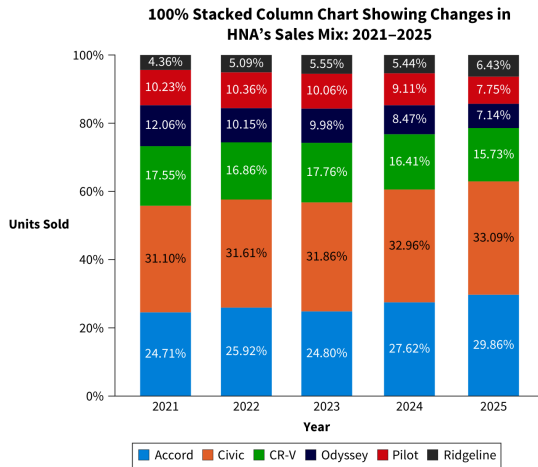


ILLUSTRATION 7.41

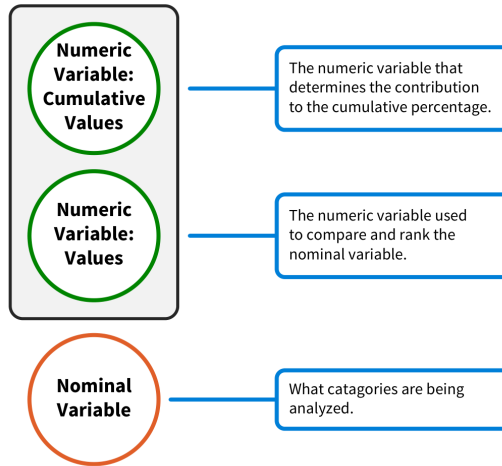
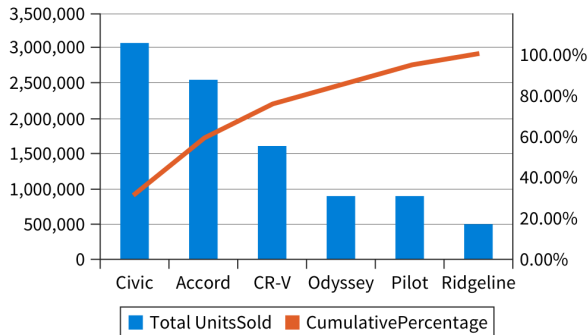


ILLUSTRATION 7.42

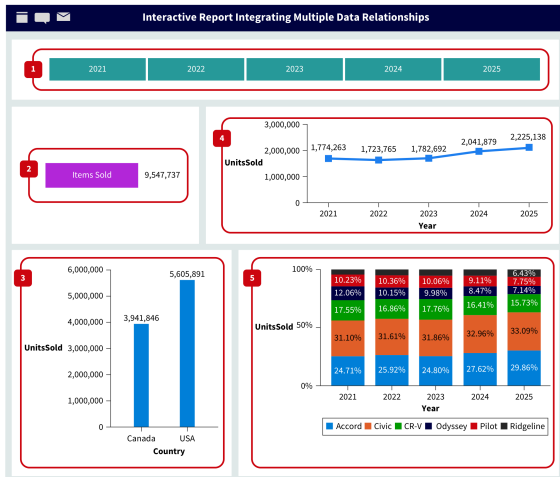
Model	TotalUnitsSold	CumulativePercentage
Civic	3,074,040	32.20%
Accord	2,555,799	58.97%
CR-V	1,603,686	75.76%
Odyssey	898,441	85.17%
Pilot	897,807	94.58%
Ridgeline	517,964	100.00%
Total	9,547,737	

(A) PivotTable



(B): Line and Column Chart

ILLUSTRATION 7.43



Bảng điều khiển (Dashboards)

- **Dashboard là gì?** Là giao diện trực quan tổng hợp nhiều biểu đồ, chỉ số quan trọng (KPIs) trên cùng một màn hình.
- **Tính Tương tác (Interactivity):** Người dùng có thể click, lọc (Slicers), khoan sâu (Drill-down) để khám phá chi tiết (Ví dụ: Từ doanh thu Tổng Công ty khoan xuống từng Chi nhánh).
- **Giá trị Kế toán:** Giúp Ban Lãnh đạo ra quyết định nhanh chóng, dựa trên dữ liệu (Data-driven decision making).

ILLUSTRATION 7.44

Slicer

1	
Filters	Columns
Year	
Rows	Values

Card

2	
Filters	Columns
Country Model Year	
Rows	Values
	TotalUnitsSold

Line chart

4	
Filters	Columns
Country Model	
Rows	Values
Year	TotalUnitsSold

Column chart

3	
Filters	Columns
Model Year	
Rows	Values
Country	TotalUnitsSold

100% Stacked column chart

5	
Filters	Columns
Country	Model
Rows	Values
Year	TotalUnitsSold

ILLUSTRATION 7.45

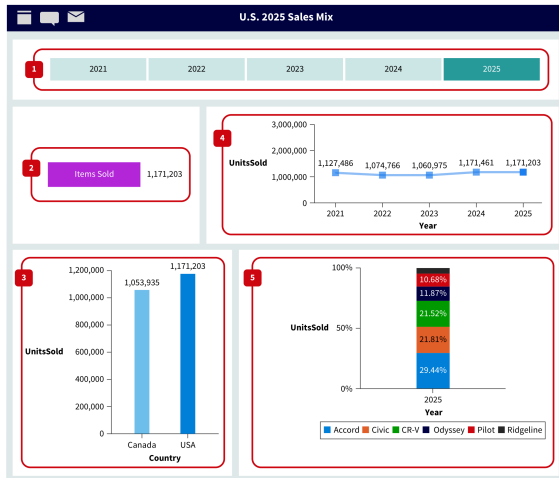


ILLUSTRATION 7.46



ILLUSTRATION 7.47

The screenshot shows the Microsoft Excel interface. The **Insert** tab is selected in the ribbon, indicated by a red circle and the number **1**. Within the **Charts** group on the Insert ribbon, the **Histogram** icon is circled with a red circle and labeled **2**. Below it, the **Box and Whisker** icon is circled with a red circle and labeled **3**. The background displays a data table with the following content:

	A	B	C	D
1	ID	Model	Profit	
2	1	CIVIC	2125	
3	2	CR-V	2798	

ILLUSTRATION 7.48

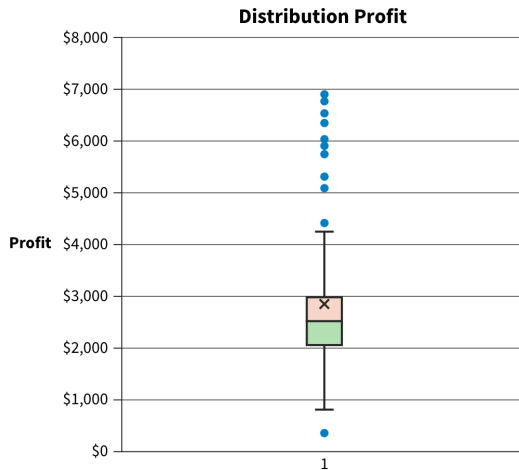


ILLUSTRATION 7.49

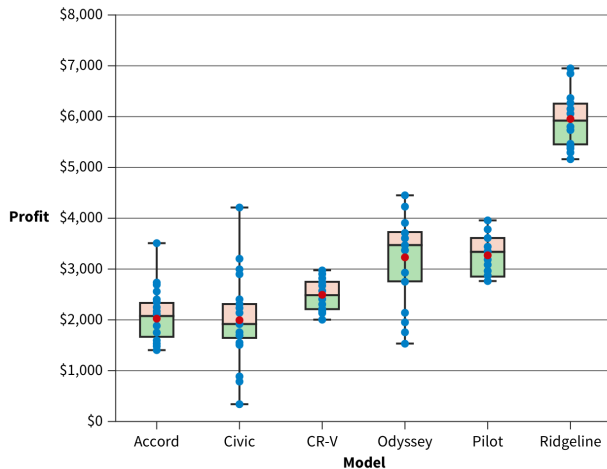


ILLUSTRATION 7.50

Model	TotalUnitsSold	CumulativePercentage
Civic	3,074,040	32.20%
Accord	2,555,799	58.97%
CR-V	1,603,686	75.76%
Odyssey	898,441	85.17%
Pilot	897,807	94.58%
Ridgeline	517,964	100.00%

ILLUSTRATION 7.51

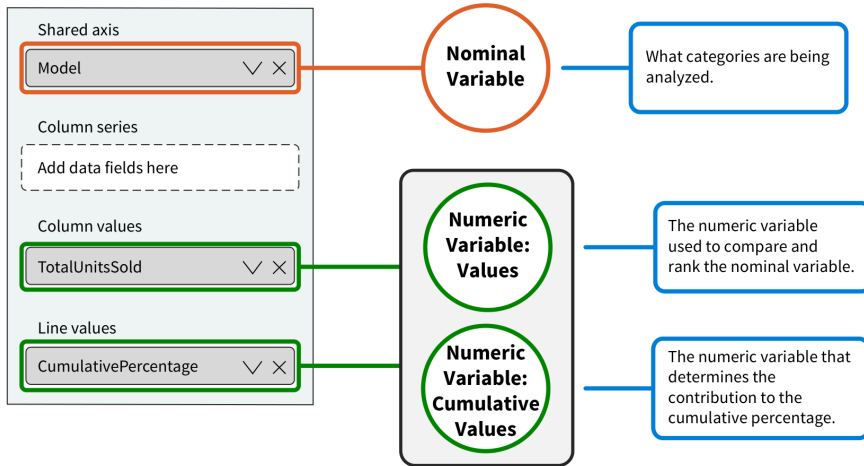
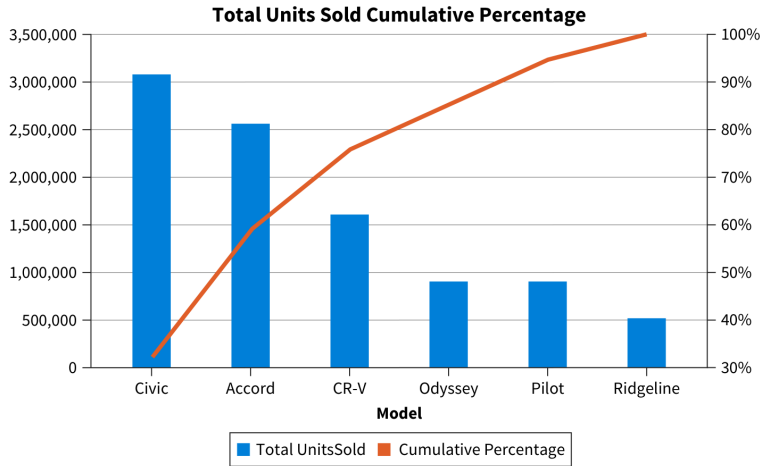


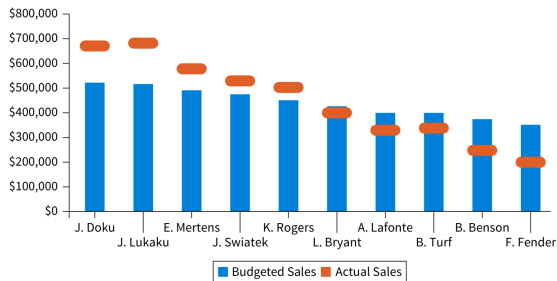
ILLUSTRATION 7.52



Phần Bài tập Ngắn

Brief Exercises (BE)

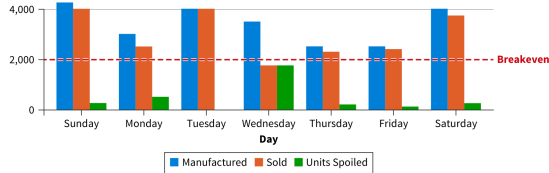
BE 7.6 & BE 7.7



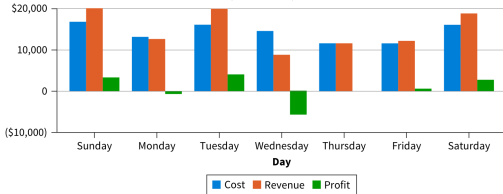
ProductLine	LastYear	ThisYear
Computers	122322	132889
Phones	789112	998325
Printers	67891	32445
TVs	322112	401223

BE 7.8 & BE 7.9

Units Manufactured, Sold, and Spoiled



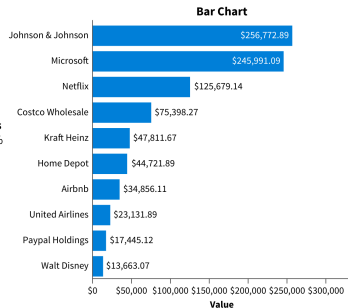
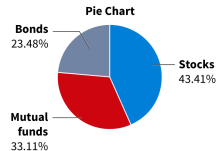
Cost, Revenue, and Profit



Account	2021	2022	2023	2024	2025
Sales	\$ 3,129,666	\$ 3,518,891	\$ 4,015,671	\$ 5,913,556	\$ 6,123,411
Accounts Receivable	\$ 425,662	\$ 436,771	\$ 491,223	\$ 549,901	\$ 618,891

BE 7.10 & BE 7.11A

Account	2021	2022	2023	2024	2025
Revenues	\$ 3,476,811	\$ 3,337,714	\$ 3,197,633	\$ 3,159,912	\$ 3,087,761
Gross Profit	\$ 1, 890,556	\$ 1,735,551	\$ 1,640,941	\$ 1,554,891	\$ 1,438,664
Net Profit	\$ 753,891	\$ 827,714	\$ 970,022	\$ 1,000,033	\$ 1,110,989



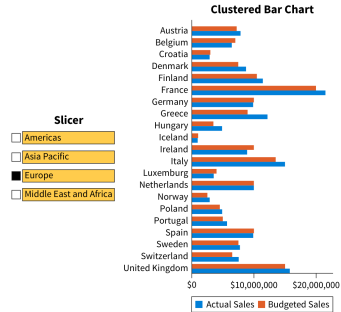
BE 7.11B & BE 7.12A

Pie Chart

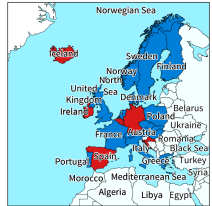
Filters	Columns
Rows	Values

Bar Chart

Filters	Columns
Rows	Values



Map



Slicer

Filters	Columns
Rows	Values

Clustered Bar Chart

Filters	Columns
Rows	Values

Map

Filters	Columns
Rows	Values

Phần Bài tập

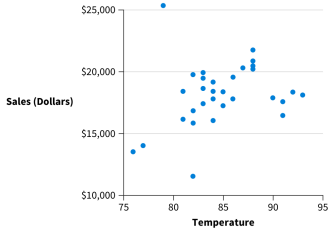
Exercises (EX)

EX 7.3 & EX 7.4

ProductLine	2024 UnitsSold	2025 UnitsSold
Adventure	2771	5741
City	8001	6799
Cyclocross	13998	14881
Family	23488	22476
Fitness	8112	11665
Mountain	7500	7430
Tandem	1089	722
Touring	3801	4022

Day	Deli	Meat	Chicken	Fish
2/1/2025	\$ 27,234	\$ 87,665	\$ 17,074	\$ 5,992
2/2/2025	\$ 34,889	\$ 43,141	\$ 38,827	\$ 10,436
2/3/2025	\$ 19,067	\$ 61,675	\$ 15,447	\$ 12,024
2/4/2025	\$ 17,812	\$ 46,855	\$ 32,029	\$ 5,285
2/5/2025	\$ 22,357	\$ 42,584	\$ 31,342	\$ 9,361
2/6/2025	\$ 11,525	\$ 55,635	\$ 42,466	\$ 9,019
2/7/2025	\$ 12,052	\$ 38,495	\$ 41,091	\$ 24,889
2/8/2025	\$ 28,554	\$ 82,009	\$ 24,739	\$ 7,563
2/9/2025	\$ 41,776	\$ 45,106	\$ 15,990	\$ 12,569
2/10/2025	\$ 16,200	\$ 44,461	\$ 25,780	\$ 11,666

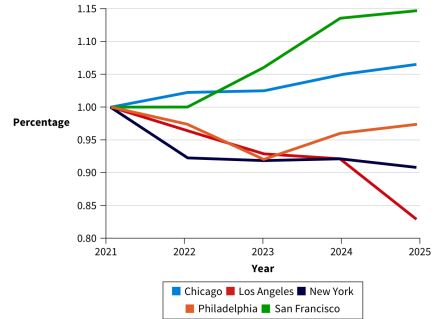
EX 7.5 & EX 7.6



Date	Temperature	Sales
7/1/25	82	\$ 16,820
7/2/25	81	\$ 16,128
7/3/25	81	\$ 18,412
7/4/25	79	\$ 25,373
7/5/25	83	\$ 17,396
7/6/25	85	\$ 17,236
7/7/25	83	\$ 18,628
7/8/25	84	\$ 18,393
7/9/25	86	\$ 17,773
7/10/25	88	\$ 21,761
7/11/25	88	\$ 20,411
7/12/25	91	\$ 17,558
7/13/25	93	\$ 18,104
7/14/25	82	\$ 11,498
7/15/25	76	\$ 13,497
7/16/25	77	\$ 13,985
7/17/25	82	\$ 15,825
7/18/25	84	\$ 16,031
7/19/25	88	\$ 20,248
7/20/25	92	\$ 18,333
7/21/25	91	\$ 16,442
7/22/25	90	\$ 17,872
7/23/25	88	\$ 20,923
7/24/25	82	\$ 19,752
7/25/25	83	\$ 19,919
7/26/25	84	\$ 19,153
7/27/25	83	\$ 19,494
7/28/25	85	\$ 18,361
7/29/25	87	\$ 20,311
7/30/25	86	\$ 19,544
7/31/25	84	\$ 17,790

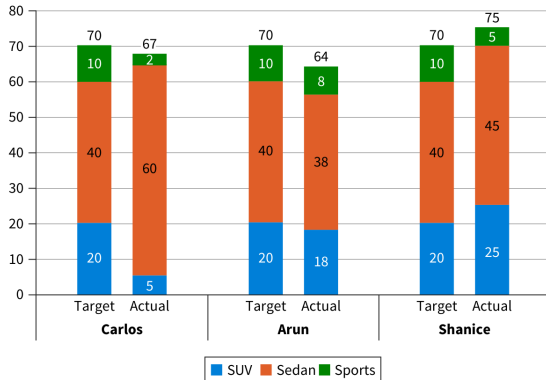
Number of Candy Trucks

Year	Chicago	Los Angeles	New York	Philadelphia	San Francisco
2021	1,240	1,712	1,350	750	920
2022	1,267	1,650	1,245	730	920
2023	1,270	1,590	1,240	690	975
2024	1,300	1,575	1,243	720	1,044
2025	1,320	1,420	1,225	730	1,055

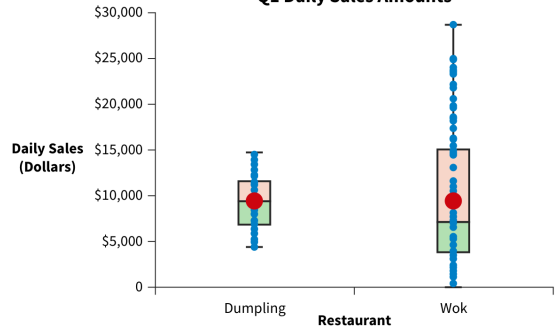


EX 7.7 & EX 7.8

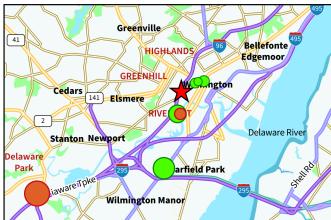
2025 Unit Sales



Q1 Daily Sales Amounts



EX 7.9 & EX 7.10A



Chain	Hotel	Location	Rate/Night
Marriott	Courtyard Wilmington Downtown	1102 N West Street, Wilmington, Delaware, 19801	127
Hilton	DoubleTree by Hilton Hotel Wilmington	4727 Concord Pike, Wilmington, DE 19803	118
Marriott	Fairfield Inn & Suites Wilmington New Castle	2117 N Dupont Hwy, New Castle, DE 19720-6308	202
Hilton	Homewood Suites by Hilton Wilmington Downtown	820 Justison St, Wilmington, DE 19801	145
Hilton	Hotel Wilmington/Christiana	100 Continental Drive Newark, Delaware 19713-4301, USA	224
Marriott	Residence Inn Wilmington Downtown	1300 North Market Street, Wilmington, Delaware 19801	142
Marriott	Sheraton Suites Wilmington Downtown	422 Delaware Avenue, Wilmington, Delaware, 19801	136
Marriott	The Westin Wilmington	818 Shipyard Drive, Wilmington, Delaware 19801	175

Warehouse	Product	QOH
Boston	M1750	412
New York	M1750	288
Philadelphia	M1750	298
Baltimore	M1750	396
Boston	M2450	49
New York	M2450	658
Philadelphia	M2450	321
Baltimore	M2450	199
Boston	M2950	420
New York	M2950	519
Philadelphia	M2950	344
Baltimore	M2950	618
Boston	M3750	400
New York	M3750	400
Philadelphia	M3750	150
Baltimore	M3750	55

EX 7.10B & EX 7.11

(A)	Filters	Columns
		Product
	Rows	Values
	Warehouse	TotalQuantity OnHand

(B)	Filters	Columns
	Product	
	Rows	Values
	Warehouse	TotalQuantity OnHand

(C)	Filters	Columns
	TotalQuantity OnHand	
	Rows	Values
	Product	TotalQuantity OnHand

AutoSave ☐ Off						
	A	B	C	D	E	F
1	Bethany Beach Ice Cream Stores Revenues 2021-2025					
2	Store	2021	2022	2023	2024	2025
3	Laureen's	\$ 283,776	\$ 309,466	\$ 340,011	\$ 418,966	\$ 498,007
4	Softer-Ice	\$ 76,000	\$ 123,000	\$ 266,128	\$ 314,000	\$ 478,326
5	YuMeeS	\$ 211,984	\$ 192,000	\$ 163,000	\$ 149,000	\$ 129,000
Sheet1						

EX 7.12 & EX 7.13

Size Total Assets	Income Subject To Tax	Total Income Tax After Credits
Zero Assets	\$ 55,884,102	\$ 11,594,836
\$1 under \$500,000	\$ 7,829,008	\$ 1,569,228
\$500,000 under \$1,000,000	\$ 4,829,204	\$ 956,103
\$1,000,000 under \$5,000,000	\$ 17,223,201	\$ 3,496,487
\$5,000,000 under \$10,000,000	\$ 10,086,326	\$ 2,015,713
\$10,000,000 under \$25,000,000	\$ 19,143,340	\$ 3,805,232
\$25,000,000 under \$50,000,000	\$ 14,384,822	\$ 2,811,835
\$50,000,000 under \$100,000,000	\$ 16,854,446	\$ 3,293,822
\$100,000,000 under \$250,000,000	\$ 25,807,956	\$ 4,772,611
\$250,000,000 under \$500,000,000	\$ 26,192,384	\$ 4,679,319
\$500,000,000 under \$2,500,000,000	\$ 126,273,758	\$ 20,835,589
\$2,500,000,000 or more	\$ 1,632,169,980	\$ 184,855,832

PENNSYLVANIA RESEARCH & DEVELOPMENT TAX CREDIT - 2021 REPORT

2020 RECIPIENTS

R&D CREDITS - DECEMBER 2020 (\$M)					
BUSINESS TYPE	APPLICANTS	% OF APPLICANTS	TENTATIVE	ACTUAL	% OF ACTUAL
SMALL	470	42.2%	\$18.1	\$11.0	20.0%
NOT SMALL	643	57.8%	\$189.7	\$44.0	80.0%
TOTAL	1,113	100.0%	\$207.7	\$55.0	100.0%

EX 7.14 & EX 7.15

Type	Source	Number
Dirty Rooms	Hotel Website	4
Dirty Rooms	Calls	22
Elevator	Calls	2
Poor Wifi	Hotel Website	4
Poor Wifi	ratemyhotel.com	2
Poor Wifi	Calls	7
Rodents	Calls	2
Room Temperature	Calls	2
Unexpected Fees	Letter	14
Unexpected Fees	Hotel Website	60
Unexpected Fees	ratemyhotel.com	16
Unexpected Fees	Calls	90
Unfriendly Staff	Letter	2
Unfriendly Staff	Hotel Website	105
Unfriendly Staff	ratemyhotel.com	44
Unfriendly Staff	Calls	45

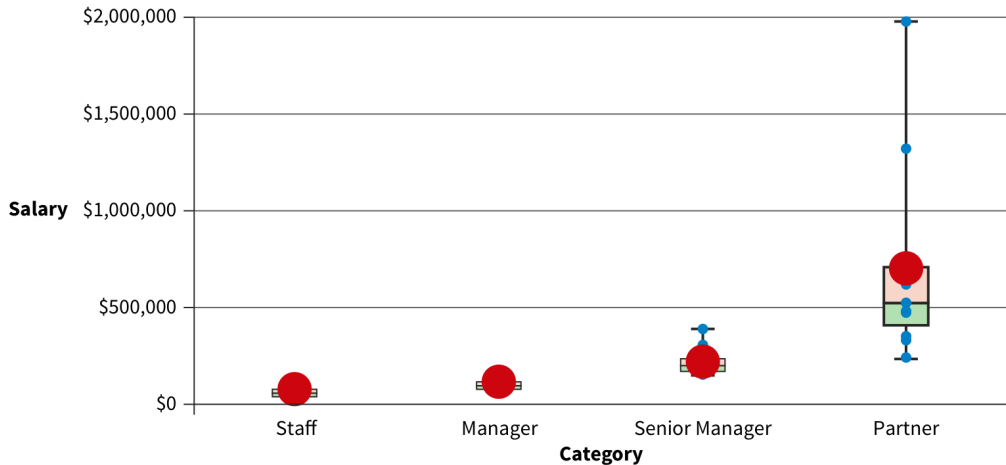
Filters	Columns
Store	
Rows	Values

Filters	Columns
Store ProductName	Store
Rows	Values
Year	TotalAmount

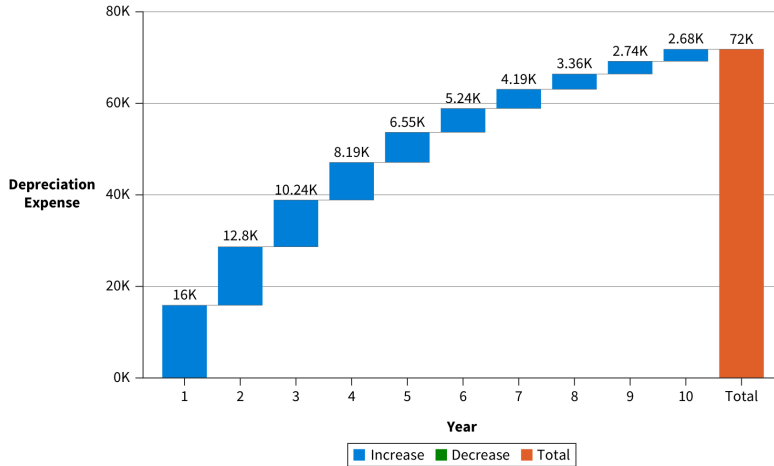
Filters	Columns
ProductName	
Rows	Values

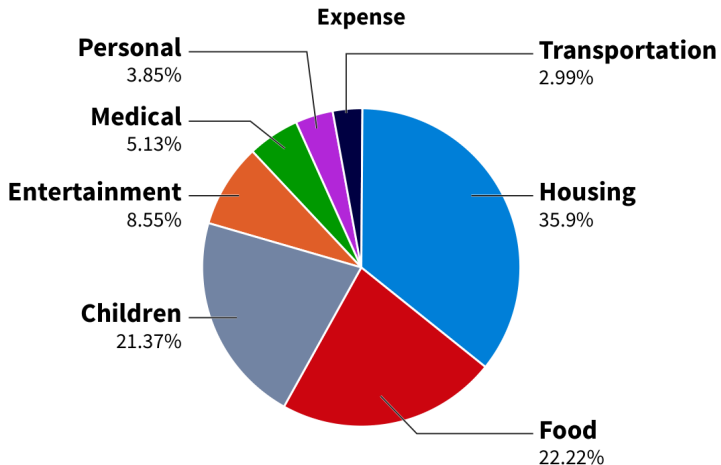
Filters	Columns
	ProductName
Rows	Values
Year	TotalAmount

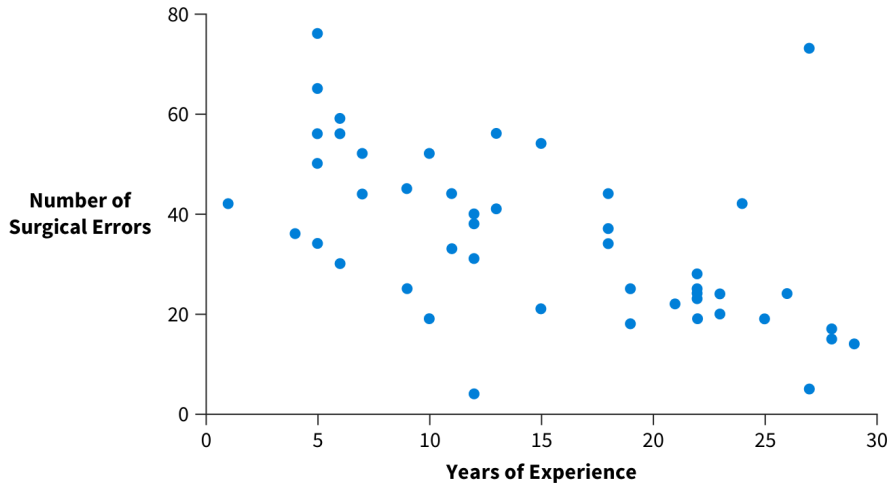
LO 2.10



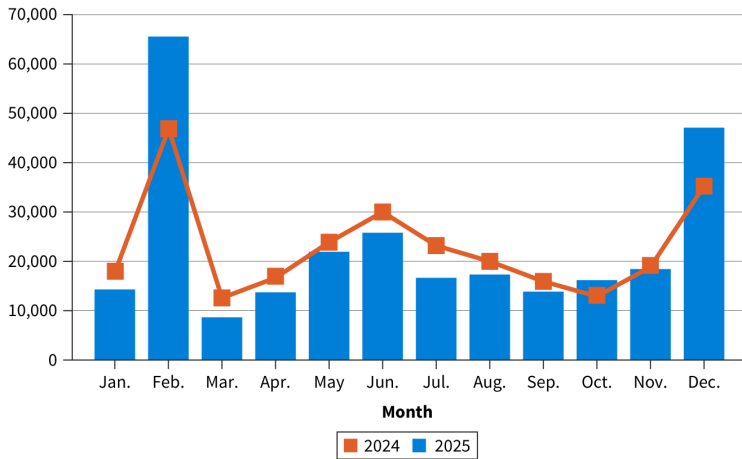
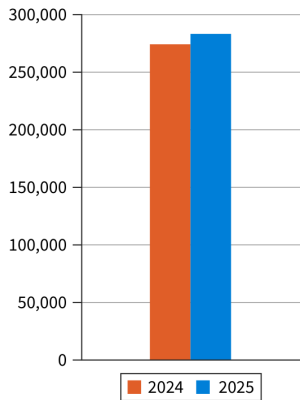
Region	2020 Profits
North-America	19,922,189
South-America	89,900,011
Western-Europe	3,008,223
Eastern-Europe	2,347,781
Africa	890,002
Asia	122,343







LO 3.15



LO 3.16

A

Filters	Columns
Country ProductType	Year
Rows	Values
ProductType	TotalRevenues

C

Filters	Columns
ProductType Year	
Rows	Values
Country Year	TotalRevenues

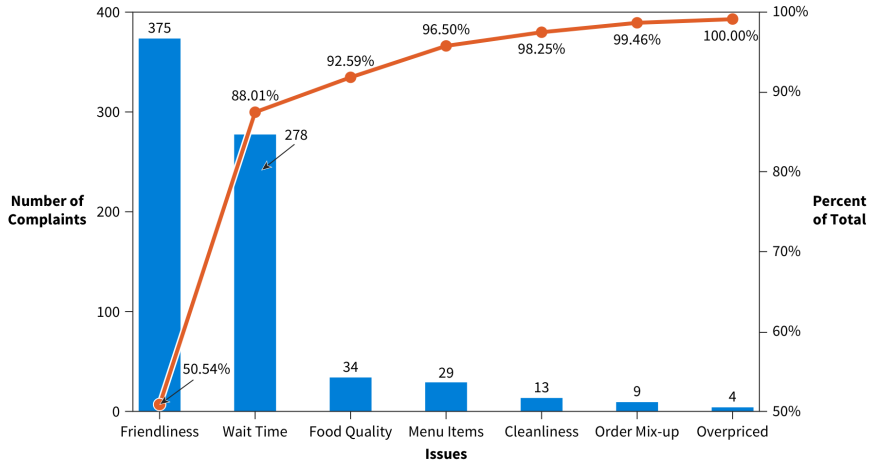
B

Filters	Columns
Country Year ProductType	Quarters
Rows	Values
ProductType	TotalRevenues

D

Filters	Columns
Country Year	Quarters
Rows	Values
ProductType	TotalRevenues

	Hair	Nails	Tanning	Totals
Revenues	\$ 123,451	\$ 37,987	\$ 55,231	\$ 216,669
Variable Costs	\$ 79,862	\$ 21,324	\$ 13,412	\$ 114,598
Fixed Costs	\$ 18,776	\$ 15,112	\$ 25,771	\$ 59,659
Profits	\$ 24,813	\$ 1,551	\$ 16,048	\$ 42,412

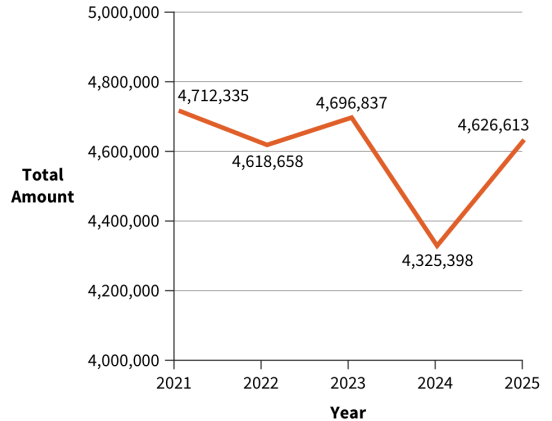
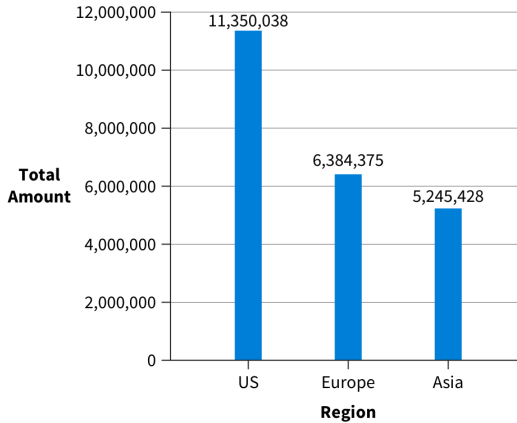


LO 3.19

Stocks 41.37%	Bonds 25.80%	Cash Equivalents 12.24%
	Cash 13.32%	Commodities 7.27%

Region	2021	2022	2023	2024	2025
Asia	789,112	977,111	1,009,112	1,125,521	1,344,572
Europe	1,256,111	1,238,771	1,398,814	1,199,765	1,290,914
U.S.	2,667,112	2,402,776	2,288,911	2,000,112	1,991,127

LO 3.20B

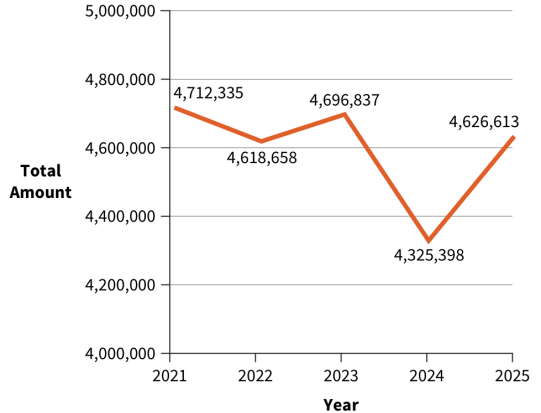
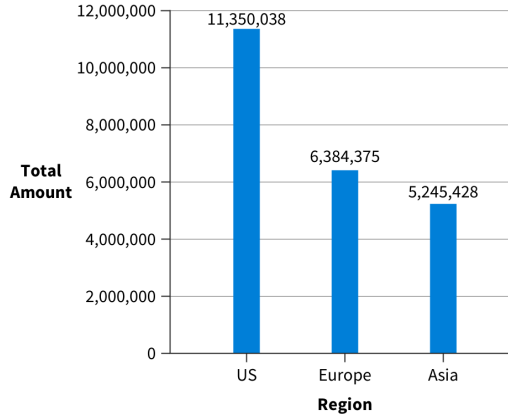


Bar Chart

Filters	Columns
Year	
Rows	Values
Region	Total Amount

Line Chart

Filters	Columns
Region	
Rows	Values
Year	Total Amount

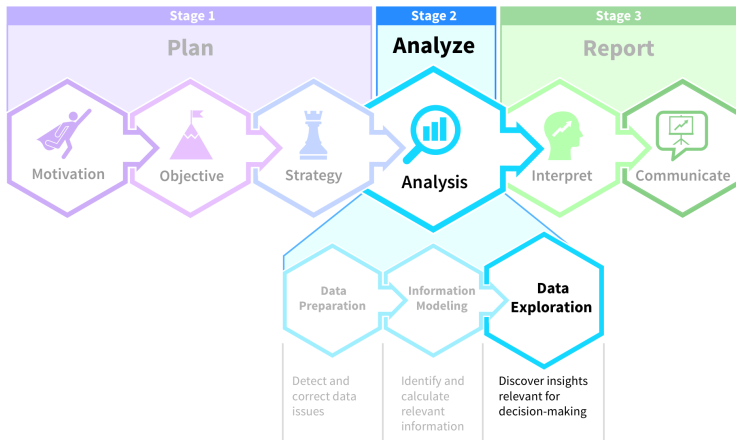


Bar Chart

Filters	Columns
Year	
Rows	Values
Region	Total Amount

Line Chart

Filters	Columns
Region	
Rows	Values
Year	Total Amount



Apply It 7.1A

PivotTable Fields

Choose fields to add to report: 

Search 

- ☐ ID
- ☐ Month
- ☒ **Product**
- ☒ **Employee**
- ☐ NumberOfDefectiveItems
- ☐ Total
- ☒ **Cost**

More Tables..

Drag fields between areas below:

Filters	Columns
	Product ▼

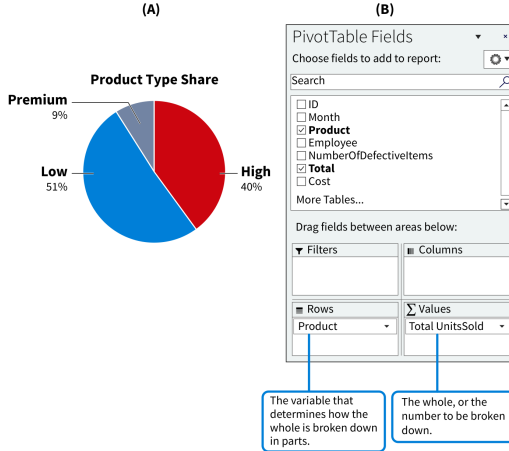
Rows	Values
Employee ▼	Cost of Defective Cans ▼

Apply It 7.1B

Cost of Defective Cans Product Type 				
Employee 	High	Low	Premium	Total
Andre	\$ 30,500	\$ 20,750	\$ 1,530	\$ 52,780
Bianca	\$ 24,750	\$ 24,250	\$ 1,147	\$ 50,147
Camille	\$ 35,500	\$ 14,750	\$ 825	\$ 51,075
Total	\$ 90,750	\$ 59,750	\$ 3,502	\$ 154,002

Apply It 7.2

Part-to-Whole Relationship with Pie Chart



Apply It 7.3A



Apply It 7.3B

Exploration Space: Happy Colors Defective Cans Analysis

Slicers

1

Filters	Columns
Month Product	
Rows	Values

100% Stacked column chart

3

Filters	Columns
Month	Product
Rows	Values
Employee	Total Number of Defective Cans

Column chart

2

Filters	Columns
Month Product	
Rows	Values
Employee	Total Number of Defective Cans

Column chart

4

Filters	Columns
Month Product	
Rows	Values
Employee	Relative Number of Defective Cans

Exploration Space: Happy Colors Defective Cans Analysis

Slicers

1

Filters	Columns
Month Product	
Rows	Values

100% Stacked column chart

3

Filters	Columns
Month Product	
Rows	Values
Employee	Total Number of Defective Cans

Column chart

2

Filters	Columns
Month Product	
Rows	Values
Employee	Total Number of Defective Cans

Column chart

4

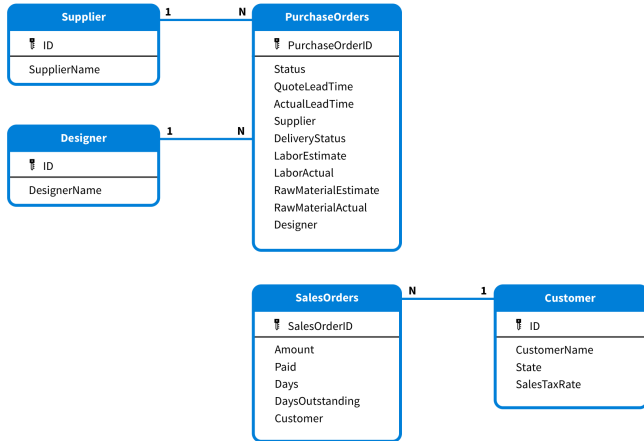
Filters	Columns
Month Product	
Rows	Values
Employee	Relative Number of Defective Cans

Apply It 7.3A_2



NoTable 7.16

Data Model



Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!